

AI USED: Chat GPT and Claude AI

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What this part is doing, in plain terms

This is the first real "prompt engineering" exercise — you're going to see, with your own eyes, why a lazy prompt gives mediocre results and a well-designed prompt gives you something actually usable. This is the whole point of the lab: it's not enough to just call an LLM, you have to *engineer* what you ask it.

You'll build two versions:

- **V1** — deliberately bad/naive, just to have a baseline to compare against
- **V2** — a properly engineered prompt with le, constraints, and structure

Write a message...

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What this section is doing, in plain terms

Up to now you've been testing the AI with generic questions (savings product names). Now you're loading the **real dataset** you'll actually work with for the rest of the lab — six fictional loan application letters from a Ghanaian microfinance context, written the way real applicants would actually write (some organized and detailed, some messy and vague — that's intentional, it mirrors real life).

Two dictionaries are being created:

1. **LETTERS** — six loan applications (**L001**  **006**), each one a block of raw text exactly as an applicant might submit it. Some are ng applications (clear numbers, collateral, track record — like L001 and L003). Some are weak or risky (vague amounts

Write a message...

So this part is checking for consistency, if the film is consistent before we move to our work

Yes, exactly — that's the core idea. You've got it right.

More precisely: it's checking **how temperature affects consistency**, so you understand you have *control* over it. At `temperature=0.0`, you're telling the model "be as consistent/predictable as possible." At `temperature=1.2`, you're telling it "feel free to vary your answers a lot."

The deeper point of this experiment is: **you now know this dial exists and what it does**, so when you build the actual loan decision-support tool next, you can *deliberately choose* the right setting instead of just leaving it at some random default.

For example:

- Summarizing a loan application factually  you'd want low temperature (consistency, no creative embellishment)

Write a message...



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so I want to understand every part in layman terms but since we have done two parts already, I would advice you to add personal explanation to it so I would know and can follow what it going on.for the part we have done already, just explain it now to me so that for the next ones you add explanation to every part I paste

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totally fair — let's recap everything in plain terms so you actually understand the "why" behind what you built, not just that it worked.

Part 0 — Setting up safely

Think of your API key like a password to a paid service. If it leaks into a public GitHub repo, anyone can steal it and rack up charges on your account (or use up your free tier). So we:

- Made a `.gitignore` file — this is like a "do not photograph" list for git. It tells git "never track this file, no matter what," so your secret key file (`.env`) never accidentally gets uploaded.

Write a message...

-

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Claude is AI and can make mistakes. Please double-check responses.

What you're looking for

Don't immediately assume your V1 is good. Read the generated summaries critically.

Ask yourself:

- Did it stay factual?
- Did it add information that wasn't in the letter?
- Did it leave out important financial information?
- Did it focus on irrelevant details?
- Did it make the applicant sound better or worse than the letter actually says?
- Is it actually **3–4 sentences**?
- Would a loan officer be able to quickly understand the application?

Step 2: Create `SUMMARY_PROMPT_V2`

Now improve the prompt by explicitly giving the model:

1. A **role**
2. A specific **task**
3. **Factuality constraints**
4. A **length constraint**
5. An instruction against hallucination/invention

You can use:



extraction labels for three of them (you will use these for Section 4).
Read at least two letters fully before moving on — you cannot engineer prompts for text you have not read.
LETTERS = {
"L001": ""Dear Sir/Madam,
My name is Akosua Mensah and I have been selling provisions at
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This section is mainly **dataset setup and reading**, so there isn't much code you need to add. The lecturer is making sure you understand the actual loan applications before you start designing prompts.

What you need to do

Run the provided cell exactly as given:

```
Python Run  
print(f"{len(LETTERS)} letters loaded.")
```

You should get:

```
6 letters loaded. Copy
```

You should also read at least two letters completely. I recommend reading **L001, L003, and L006**, because these are the three letters that have the **gold-standard labels** and will later be used to evaluate your LLM's extraction.

Yes — this is **Lab 4**, and the part you've pasted is asking you to complete **Part 0** and **Part 1.1–1.2**. I'll walk you through exactly what to put in the notebook, including the code and what to write under each reasoning question.

One important thing first: **do not put your actual API key in the notebook or in the code you commit to GitHub**.

Part 0 — API Key Setup

Since the lab specifically gives you the Colab Secrets option, if you're using **Google Colab**, I recommend that method.

1. Add your API key to Colab Secrets

In Colab:

Left sidebar →  **Secrets** → **Add a new secret**

Use:

```
GROQ_API_KEY Copy
```

and paste your Groq API key as the value.

Then make sure **Notebook access** is enabled.

2. API-key setup cell



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What the dataset contains

There are six loan application letters:

ID	Applicant	Loan	Main purpose
L001	Akosua Mensah	GHS 8,000	Deep freezer/frozen foods
L002	Kwame Boateng	GHS 25,000	Trotro repair/debts